
Depressive Rumination

NATURE, THEORY AND TREATMENT

Edited by

COSTAS PAPAGEORGIOU

University of Lancaster, UK

ADRIAN WELLS

University of Manchester, UK



WILEY

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About the Editors

Costas Papageorgiou is Senior Lecturer at Lancaster University and was Deputy Director of the Specialist Service for Affective (Mood) Disorders in Manchester. He obtained a BSc from the University of Buckingham and an MA and a Doctorate in Clinical Psychology from the University of Liverpool. Dr Papageorgiou has expertise in the assessment and treatment of depression. He has been extensively involved in investigating rumination and depression and has co-authored the first empirical studies examining the link between rumination, depression, and metacognition. Dr Papageorgiou has presented papers in numerous national and international conferences and published peer-reviewed articles on cognitive and metacognitive theory and therapy of emotional disorders, particularly depression. He has guest-edited two special issues on the Persistence of Depressive and Anxious Thinking (*Behavior Therapy*) and Rumination and Depression (*Cognitive Therapy and Research*).

Adrian Wells is Reader in Clinical Psychology at the University of Manchester and Professor at the Norwegian University of Science and Technology, Trondheim. He is recognized for his contribution to the development of cognitive theory and therapy of emotional disorders. He has published widely in peer-review journals and has authored/co-authored several groundbreaking books in the field. These works include: *Cognitive Therapy of Anxiety Disorders: A Practice Manual and Conceptual Guide* (John Wiley & Sons); *Emotional Disorders and Metacognition: Innovative Cognitive Therapy* (John Wiley & Sons); and *Attention and Emotion: A Clinical Perspective* (Lawrence Erlbaum). Professor Wells' research interests are in the areas of cognitive–attentional processes in emotional disorder, metacognition and cognitive therapy, and specific disorders including social phobia, generalized anxiety, PTSD, and obsessional problems. He is a pioneer of “metacognitive therapy” for emotional disorders.

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Preface

Depression is the most common psychological disorder incurring significant personal, social, and economic costs. Cognitive approaches have been highly influential in the conceptualization and treatment of depression. Several cognitive processes have been implicated in the development, maintenance, and recurrence/relapse of depression. In the past 15 years, persistent, recyclic, negative thinking, in the form of rumination, has attracted increasing theoretical and empirical attention as an important factor.

This book brings together leading theorists, researchers, and clinicians working in the field of rumination and depression. The aim of the book is to provide a comprehensive analysis of the nature, effects, measurement, and treatment of rumination. This volume is divided into three parts.

In Part I, five chapters describe the nature and consequences of rumination. In Chapter 1, Papageorgiou and Wells examine concepts of rumination and compare this construct with other related cognitive processes, such as worry. The chapter discusses possible functions of rumination and the relationships between rumination, depression, and metacognitive beliefs. Lyubomirsky and Tkach (Chapter 2) describe in detail a ruminative style of responding to depressed mood and review experimental and correlational research documenting its many adverse consequences. In Chapter 3, Spasojevic, Alloy, Abramson, MacCoon, and Robinson describe their research focusing on reactive rumination, which they view as a response to negative mood, negative life events, or both. Wenzlaff (Chapter 4) examines rumination in the context of mental control in which cognitive biases and thought suppression can intensify negative thinking. The deleterious impact of depressed mood on mental control is also considered as contributing to rumination. In Chapter 5, Siegle and Thayer review the physiological correlates of depressive rumination as a way of measuring and understanding its nature and underlying mechanisms. An important advantage of physiological assessments is that they may provide a more objective index of rumination.

In Part II, three leading theories of rumination are described. Nolen-Hoeksema (Chapter 6) presents her response styles theory, in which rumination is a process whereby individuals focus on the causes and consequences of depression. This, in turn, is thought to contribute to the perpetuation and exacerbation of depressive symptoms. In Chapter 7, Matthews and Wells present their information processing model of emotional disorders to account for the causes and consequences of depressive rumination. In this

multilevel model, rumination is seen as a maladaptive coping strategy closely linked to individuals' metacognitive beliefs. Martin, Shrira, and Startup (Chapter 8) review the goal progress theory of rumination and consider rumination as a function of goal progress, stop rules, and cerebral lateralization. Part II of the book concludes with a comparison and appraisal of these theories of rumination. Here Brotman and DeRubeis (Chapter 9) analyse the similarities and differences between theories and suggest areas of synthesis.

In Part III, the measurement and treatment of rumination is considered. Luminet (Chapter 10) describes specific self-report measures of rumination and related constructs and discusses their psychometric properties. In Chapter 11, Purdon presents a comprehensive review of existing psychological treatments for rumination across disorders. This chapter considers the applicability of these treatments to rumination in depression. McMillan and Fisher (Chapter 12) provide a description of common techniques used to treat depressive thinking in the course of cognitive therapy. These techniques draw on traditional strategies aimed at modifying the content of negative thoughts and beliefs in depression. In the final chapter, Wells and Papageorgiou (Chapter 13) present a new treatment approach that is specifically targeted at the depressive ruminative style of thinking. The chapter describes an array of metacognitive treatment strategies derived from an information processing model.

We have attempted to produce a book that as a body of work will help readers to identify rumination, distinguish it from related processes, understand its antecedents and consequences, and guide clinicians in their assessment and choice of treatment strategies. We hope that this volume will serve to stimulate future research on the link between rumination and depression.

Costas Papageorgiou
Adrian Wells

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I Nature and Consequences of Rumination



1 Nature, Functions, and Beliefs about Depressive Rumination

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Consider the following questions: What is rumination? How does rumination overlap with, and differ from, other cognitive processes and products? What is the role of rumination in depression? What factors are responsible for initiating and maintaining rumination, and how is rumination linked to depression? In this chapter, we address each of these questions by exploring the phenomenology of depressive rumination. The chapter begins by examining definitions of rumination. The second section reviews studies comparing depressive rumination with other forms of repetitive negative thinking. The next section considers the functions of rumination in depression. The final section explores the relationships between rumination, depression, and metacognitive beliefs.

DEFINITIONS OF RUMINATION

Rumination, crudely defined as persistent, recyclic, depressive thinking, is a relatively common response to negative moods (Rippere, 1977) and a salient cognitive feature of dysphoria and major depressive disorder. Examples of ruminative thoughts include: “why am I such a loser?”, “my mood is so bad,” “why do I react so negatively?”, “I just can’t cope with anything,” and “why don’t I feel like doing anything?” A chain of ruminative thoughts may be symptomatic of dysphoria or clinical depression, but it may also be perceived as serving a function. In view of the potential to advance our knowledge of the mechanisms of depressive onset, maintenance, and recurrence, rumination has attracted increasing theoretical and empirical interest in the past 15 years.

An important starting point in the process of understanding ruminative thinking and its link to depression is to examine notions of the concept of rumination. A number of definitions have been proposed from various psychological perspectives. According to Martin and Tesser (1989, 1996) rumination is a generic term that refers to several varieties of recurrent thinking. That is, rumination refers to the entire class of thought that has a tendency to recur. Martin and Tesser (1996, p. 7) proposed the following definition of rumination:

Rumination is a class of conscious thoughts that revolve around a common instrumental theme and that recur in the absence of immediate environmental demands requiring the thoughts. Although the occurrence of these thoughts does not depend on direct cueing by the external environment, indirect cueing by the environment is likely given the high accessibility of goal-related concepts. Although the external environment may maintain any thought through repeated cueing, the maintenance of ruminative thoughts is not dependent upon such cueing.

Nolen-Hoeksema and colleagues have been instrumental in advancing our knowledge of ruminative thinking in depression. The response styles theory of depression (Nolen-Hoeksema, 1991) conceptualizes rumination as repetitive and passive thinking about symptoms of depression and the possible causes and consequences of these symptoms. According to this perspective, rumination involves “repetitively focusing on the fact that one is depressed; on one’s symptoms of depression; and on the causes, meanings, and consequences of depressive symptoms” (Nolen-Hoeksema, 1991, p. 569).

More recent definitions of rumination have been proposed by investigating rumination on current feelings of sadness or “rumination on sadness” (Conway, Csank, Holm, & Blake, 2000) and rumination about negative inferences following stressful life events or “stress-reactive rumination” (Alloy et al., 2000; Robinson & Alloy, 2003). In Conway et al.’s (2000) definition, rumination “consists of repetitive thoughts concerning one’s present distress and the circumstances surrounding the sadness” (p. 404). According to this definition, the ruminative thoughts (1) relate to the antecedents or nature of negative mood, (2) are not goal-directed and do not motivate individuals to make plans for remedial action, and (3) are not socially shared while individuals are engaged in rumination.

Grounded on the hopelessness theory of depression (Abramson, Metalsky, & Alloy, 1989) and Beck’s (1967) cognitive theory of depression, Alloy and colleagues (Alloy et al., 2000; Robinson & Alloy, 2003) proposed a conceptual extension of Nolen-Hoeksema’s (1991) response styles theory (see also Zullow & Seligman, 1990 for a similar extension). In this extension, Alloy and colleagues developed the concept of stress-reactive rumination to refer to the tendency to ruminate on negative inferences following stressful life events. Here stress-reactive rumination is thought to occur prior to the onset of depressed mood, whereas emotion-focused rumination, as suggested by

Nolen-Hoeksema (1991), is thought to occur in response to depressed mood. Data from the Temple-Wisconsin Cognitive Vulnerability to Depression Project (Alloy & Abramson, 1999) suggest that stress-reactive rumination plays a crucial role in the aetiology of depression. Alloy et al. (2000) demonstrated that the interaction between negative cognitive styles and stress-reactive rumination predicted the retrospective lifetime rate of major depressive episodes as well as hopelessness depressive episodes. In a subsequent study, Robinson and Alloy (2003) found that the same interaction predicted the prospective onset, number, and duration of major depressive and hopelessness depressive episodes (for further details, see Chapter 3).

The review of definitions of rumination indicates that, although there are similarities between the various definitions proposed, different theorists define rumination somewhat differently. As noted by Siegle (2000), this problem is clearly reflected in existing measures of rumination. Siegle (2000) investigated the extent to which different measures of rumination represented a single construct in a factor analytic study. The results suggested that there were several separate constructs represented in the measures (for further details, see Chapter 5). Therefore, there appears to be a range of subcomponents of rumination, and it is conceivable that their contribution to dysphoria or depression may differ. Future research on rumination should clearly operationalize the type as well as components of rumination being examined.

COMPARISONS OF DEPRESSIVE RUMINATION WITH OTHER COGNITIVE PROCESSES AND PRODUCTS

Given the above conceptualizations of rumination, there are apparent similarities and differences between this process and other related cognitive processes and products (namely, negative automatic thoughts, self-focused attention or private self-consciousness, and worry). An examination of the overlap and differences between rumination and other cognitive constructs may assist in refining the concept of rumination. However, to date little is known about the similarities and differences between rumination and other cognitive constructs, or whether the similarities or differences are important contributors to psychopathology. This section reviews the literature on the overlapping and distinct features of rumination and other related or similar constructs.

RUMINATION VS. NEGATIVE AUTOMATIC THOUGHTS

Rumination may be distinguished from the negative automatic thoughts that are typical of depression. According to Beck's (1967, 1976) content specificity hypothesis, depression is characterized by thoughts containing themes of past personal loss or failure. Papageorgiou and Wells (2001a) have argued that,

although negative automatic thoughts are relatively brief shorthand appraisals of loss and failure in depression, rumination consists of longer chains of repetitive, recyclic, negative, and self-focused thinking that may well occur as a response to initial negative thoughts. Studies have also demonstrated that ruminative thinking predicts depression over and above its shared variance with several types of negative cognitions (e.g., Nolen-Hoeksema, Parker, & Larson, 1994; Spasojevic & Alloy, 2001).

RUMINATION VS. SELF-FOCUSED ATTENTION AND PRIVATE SELF-CONSCIOUSNESS

A conceptual distinction can be made between ruminative thinking and the depressive self-focusing style (Pyszczynski & Greenberg, 1987). Although the focus of the depressive style is on reducing discrepancies between ideal and real states following failure (Pyszczynski, Greenberg, Hamilton, & Nix, 1991), the focus of rumination is more specific and has been hypothesized to involve coping in the form of problem-solving, which does not necessarily occur following failure (Wells & Matthews, 1994). Rumination may also be differentiated from private self-consciousness (Fenigstein, Scheier, & Buss, 1975), a disposition to chronically self-focus and self-analyse regardless of mood. Nolen-Hoeksema and Morrow (1993) demonstrated that, although rumination remained a significant predictor of depressed mood after statistically controlling for private self-consciousness, private self-consciousness was not a significant predictor of depression after controlling for rumination. In addition to these distinctions, Papageorgiou and Wells (2001a) suggested that, while rumination in depression is likely to involve self-relevant chains of negative thoughts, not all forms of ruminative thinking are necessarily self-relevant. For instance, individuals may ruminate about the humanitarian effects of recent warfare. We believe that depressive rumination specifically encompasses self-focused thinking and negative appraisals of the self, emotions, behaviours, situations, life stressors, and coping. Thus, self-focus is a component of rumination that links to some, but not all, aspects of the content or form that rumination takes.

RUMINATION VS. WORRY

Rumination appears to be closely related to worry. Although worry is a common cognitive feature of anxiety disorders and a cardinal feature of generalized anxiety disorder, it has been reported to be elevated in individuals with depression (Starcevic, 1995). Worry has been defined as “a chain of thoughts and images, negatively affect-laden and relatively uncontrollable; it represents an attempt to engage in mental problem-solving on an issue whose outcome is uncertain but contains the possibility of one or more negative outcomes” (Borkovec, Robinson, Pruzinsky, & DePree, 1983, p. 10). Earlier research

exploring the nature of depressive and anxious thinking showed that these types of cognitions were clearly distinct phenomena (Clark & de Silva, 1985; Clark & Hemsley, 1985). The content of chains of anxious (worrisome) thoughts is likely to differ from depressive (ruminative) thoughts in that the former may be particularly characterized by themes of anticipated threat or danger in the future (Beck, 1967, 1976; Borkovec et al., 1983), while rumination may involve themes of past personal loss or failure (Beck, 1967, 1976). In a content analysis of naturally occurring worrisome thoughts, Szabo and Lovibond (2002) found that 48% of worrisome thoughts could be characterized as reflecting a problem-solving process, 17% as anticipation of future negative outcomes, 11% “rumination”, and 5% as reflecting “palliative” thoughts and “self-blame”. In another study, worrisome thinking was characterized by more statements implying catastrophic interpretations of future events than dysphoric ruminative thinking (Molina, Borkovec, Peasley, & Person, 1998). These studies suggest that there are content differences between rumination and worry.

Earlier approaches to understanding the nature of different styles of thinking had focused predominantly on the thematic content of thought in depression and anxiety. More recent theoretical and empirical evidence suggests that other dimensions of thinking, apart from content, are involved in vulnerability to, and maintenance of, psychopathology. Wells and Matthews (1994) argue that it is not only the content of perseverative negative thinking that may be relevant to understanding psychopathology but also the nature, flexibility, and beliefs about thinking that have consequences for information processing and self-regulation. According to Wells and Matthews (1994), two components of thinking styles should be considered in this context: (1) process dimensions (e.g., attentional involvement, dismissability, distraction, etc.), and (2) metacognitive dimensions (e.g., beliefs or appraisals about thinking and ability to monitor, objectify, and regulate thinking). Therefore, the study of dimensions of thinking styles may allow us to systematically construct a profile of the constituents of thinking processes that contribute to specific and/or general manifestations of psychological disturbance. To date, two studies have explored the process and metacognitive dimensions of rumination and worry (Papageorgiou & Wells, 1999a, b).

In a preliminary study, Papageorgiou and Wells (1999a) compared the process and metacognitive dimensions of naturally occurring depressive (ruminative) thoughts and anxious (worrisome) thoughts in a non-clinical sample. Participants were provided with a diary for recording and rating the content of their first and second depressive and anxious thoughts occurring during a two-week period. The results revealed that, although ruminative and worrisome thinking shared a number of similarities, they also differed on several dimensions. Figure 1.1 illustrates the differences between rumination and worry on the dimensions assessed. In comparison with rumination, worry was found to be significantly greater in verbal content, associated with more compulsion to

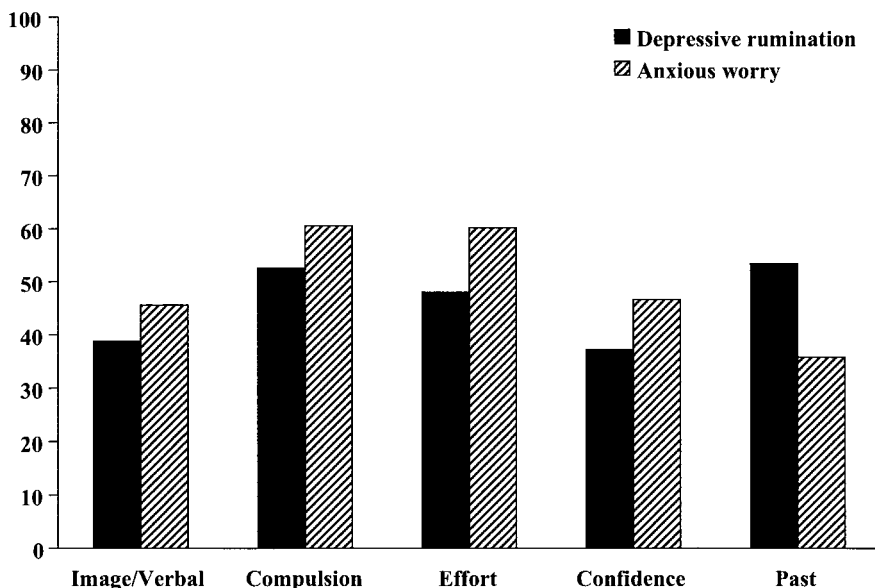


Figure 1.1. Process and metacognitive differences between depressive rumination and anxious worry in a non-clinical sample.

act, and more effort and confidence in problem-solving. Rumination was significantly more past-oriented than worry. Following adjustments for multiple comparisons, the only remaining significant differences were those concerned with dimensions of effort to problem-solve and past orientation. Relationships between dimensions of thinking and affective responses for each style of thinking were also explored in this study. This was achieved by partialling out anxiety when examining correlates of depression intensity and partialling out depression when examining correlates of anxiety intensity. Greater depression was significantly correlated with lower confidence in problem-solving ability and greater past orientation of the ruminative thoughts. In relation to the worrisome thoughts, greater anxiety was significantly correlated with less dismissibility of worry, greater distraction by worry, metaworry (i.e., worry about worry: Wells, 1994), compulsion to act on worry, and more attentional focus on worries. Therefore, these preliminary data appear to be consistent with the notion that different components of thinking style are associated with emotional disturbance (Wells & Matthews, 1994). However, the generalizability of these findings is limited by the non-clinical sample recruited.

In a subsequent study, we set out to extend these findings in clinical samples (Papageorgiou & Wells, 1999b). For this purpose, individuals whose predominant style of thinking is characterized by depressive rumination (e.g., individuals with major depressive disorder) and anxious worry (e.g., individuals with panic disorder) were recruited into the study. To reduce the overlap of

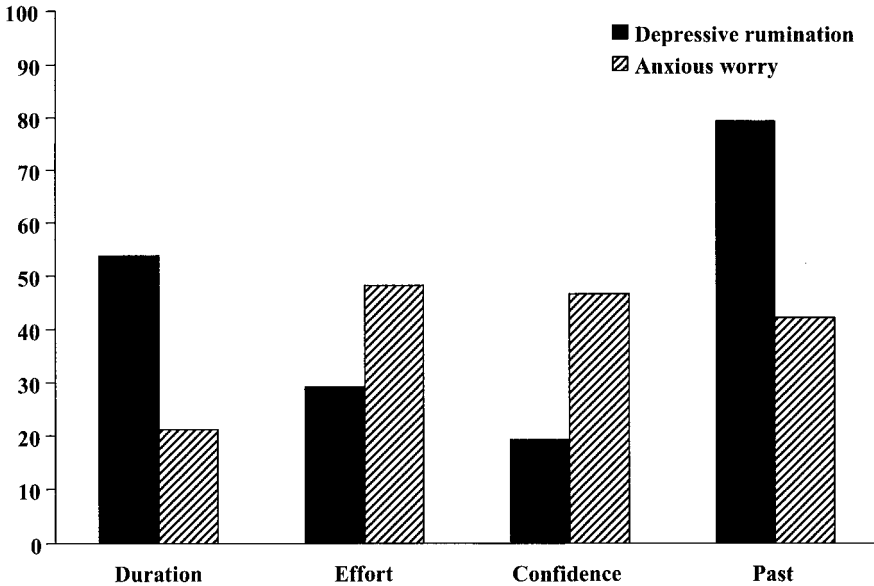


Figure 1.2. Process and metacognitive differences between depressive rumination and anxious worry in individuals with major depressive disorder.

rumination and worry, it was ensured that there was no diagnostic overlap between the two clinical samples. A non-clinical group was included in order to control for “pathological” status. We assumed that a non-clinical group would show non-pathological varieties of rumination and worry, thus enabling us to identify differences between normal and abnormal thinking styles. In this study, we aimed to address three fundamental questions. In the first question, we set out to determine whether process and metacognitive dimensions distinguish between the rumination and worry of individuals with major depressive disorder. The data showed that, in comparison with worry, the rumination of the depressed group was rated as significantly longer in duration, lower in effort to problem-solve, lower in confidence in problem-solving, and greater in past orientation. These data are presented in Figure 1.2. Following adjustments for multiple comparisons, the only remaining significant differences were those concerned with dimensions of confidence in problem-solving and past orientation.

In the second question, the objective was to establish similarities and differences between the predominant styles of pathological thinking in each disorder (i.e., rumination in major depressive disorder vs. worry in panic disorder). In comparison with the worry of the panic disorder group, the rumination of the depressed group was rated as significantly longer in duration, less controllable, less dismissible, and associated with lower effort to problem-solve, lower confidence in problem-solving, and a greater past orientation. These data are

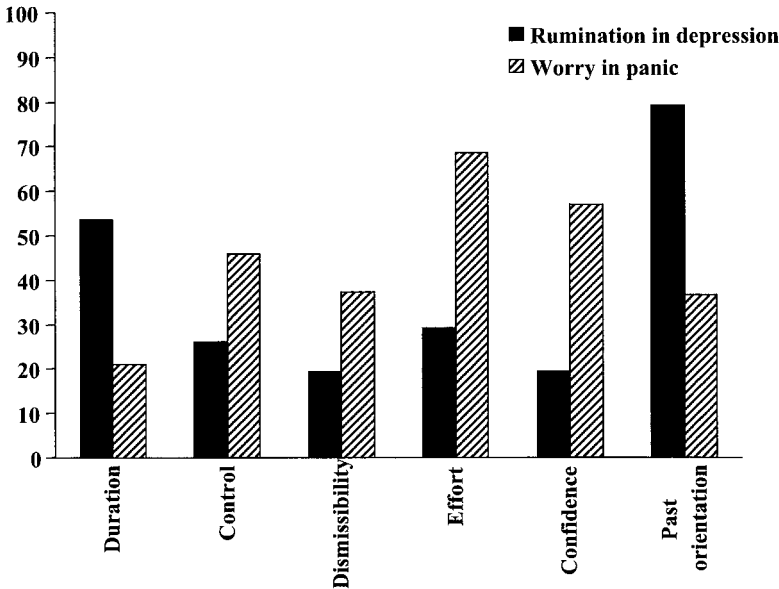


Figure 1.3. Process and metacognitive differences between rumination in patients with depression and worry in patients with panic disorder.

illustrated in Figure 1.3. Nonetheless, after adjustments for multiple comparisons, the only remaining significant differences were those concerned with dimensions of effort to problem-solve, confidence in problem-solving, and past orientation.

Finally, we addressed the question of whether dimensions of rumination differ across disorders (i.e., is pathological rumination in depression different from that in panic disorder patients and non-clinical samples whose rumination is less problematic?). The analyses demonstrated that, in comparison with the rumination of panic patients and non-clinical participants, that of the depressed group was rated as more intrusive, comprising greater metaworry, and associated with lower effort and less confidence in problem-solving, and a greater past orientation. The duration of rumination in both the depressed and panic disorder groups was significantly longer than that in the non-clinical sample. Moreover, the depressed group paid significantly more attention to their ruminative thinking than did the non-clinical sample. These data are shown in Figure 1.4. Following statistical adjustments, only the duration of rumination in the depressed group remained significantly longer than that in non-clinical participants. Thus, empirical evidence suggests that although rumination and worry share a number of similarities, they also differ on several dimensions (Papageorgiou & Wells, 1999a, b). The most reliable differences found between these two styles of thinking are effort and confidence in

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